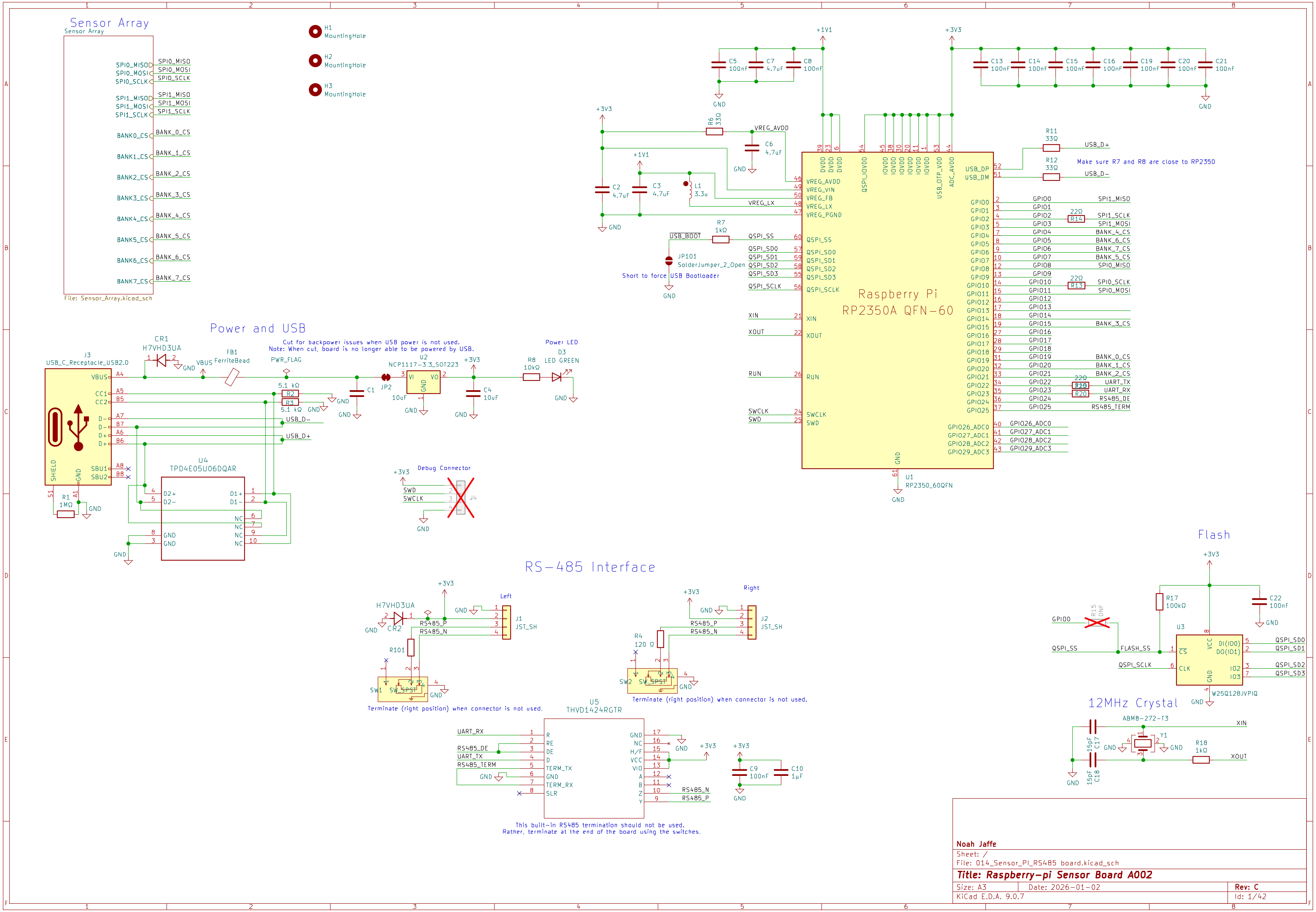


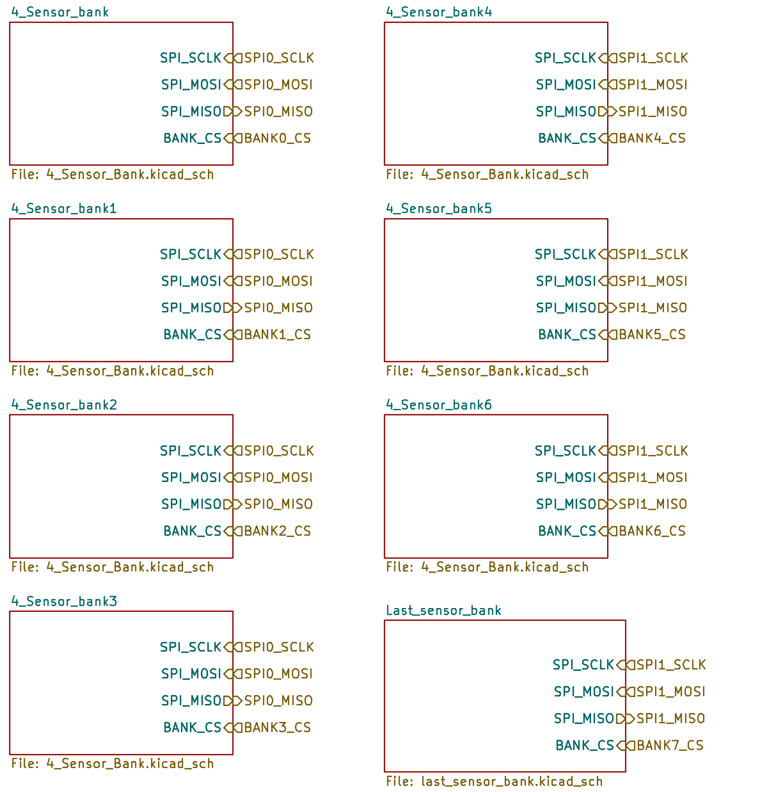
Noah Jaffe
Sheet: /
File: 014_Sensor_PI_RS485 board.kicad_sch

Title: Raspberry-pi Sensor Board A002

Size: A3 | Date: 2026-01-02 | Rev: C
KiCad E.D.A. 9.0.7 | Id: 1/42

The (up-to) 32-sensor array consists of banks of 4.

Bank 0: Sensors 0-3



Note: This sensor bank does *not* inherit from 4_Sensor_bank
This is to allow for DNP sensors, so the total number of sensors
can be a number not modulus 4

SPI Sensor Bank Connections

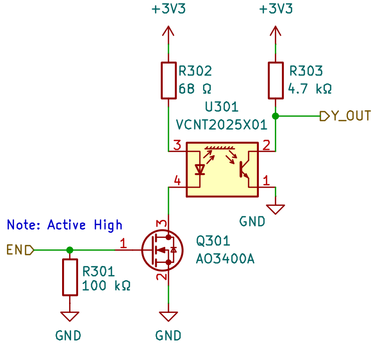
Sheet: /Sensor Array/
File: Sensor_Array.kicad_sch

Title: 8x4-Channel LED Sensor Array

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 2/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



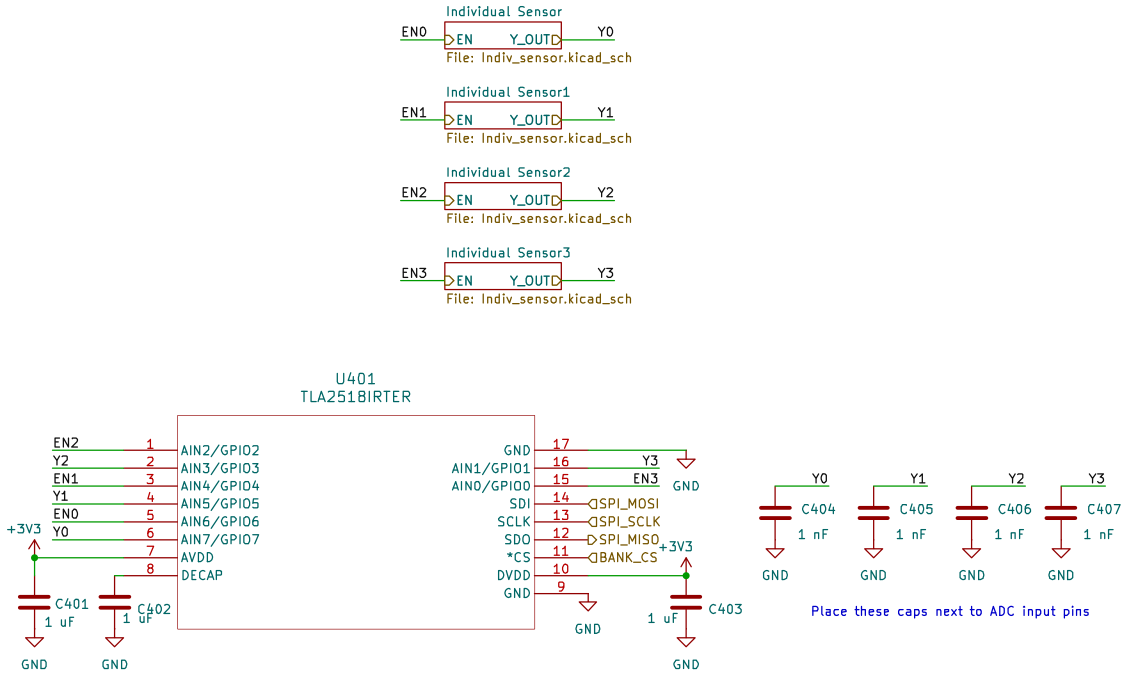
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 3/42



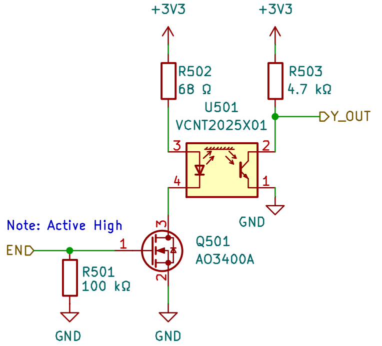
Sheet: /Sensor Array/4_Sensor_bank/
File: 4_Sensor_Bank.kicad_sch

Title: 4 to 1 Channel Sensor Array

Size: A4 | Date: 2026-01-02 | **Rev: C**
KiCad E.D.A. 9.0.7 | Id: 4/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

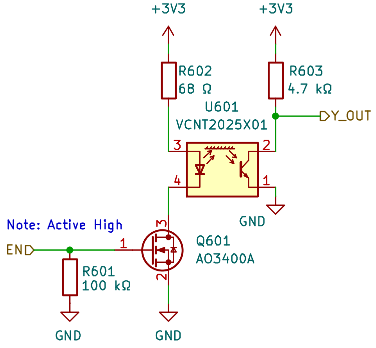
Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 5/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

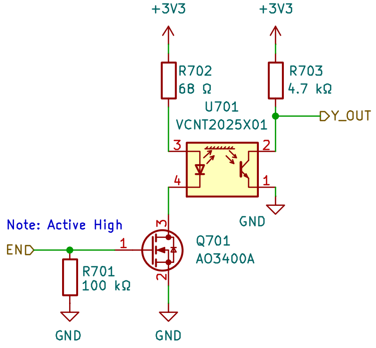
Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 6/42

łote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



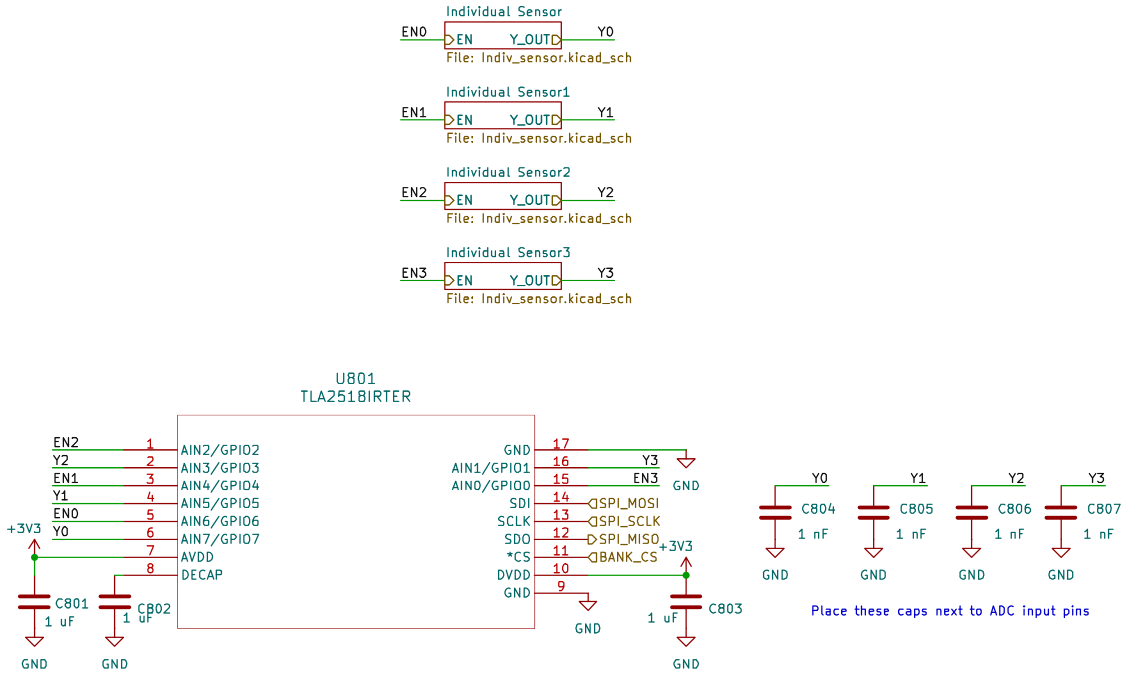
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 7/42



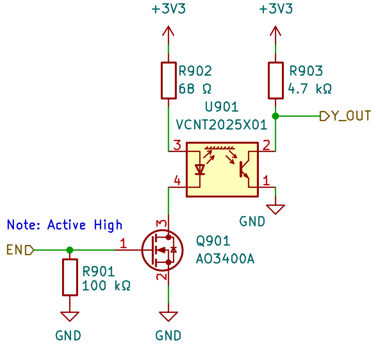
Sheet: /Sensor Array/4_Sensor_bank1/
File: 4_Sensor_Bank.kicad_sch

Title: 4 to 1 Channel Sensor Array

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 8/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 9/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

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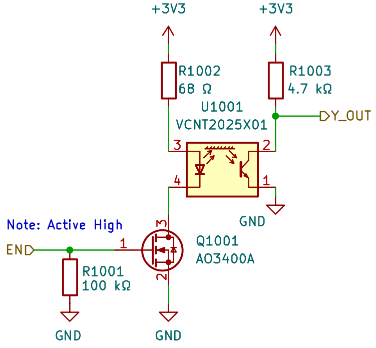
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

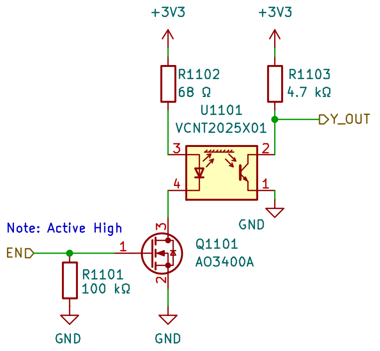
Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 10/42

ote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

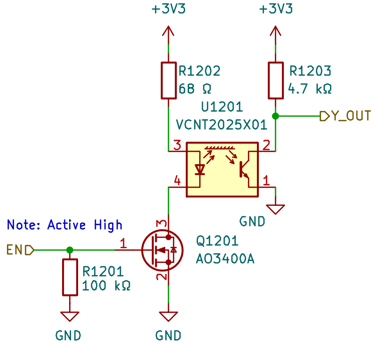
Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 11/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



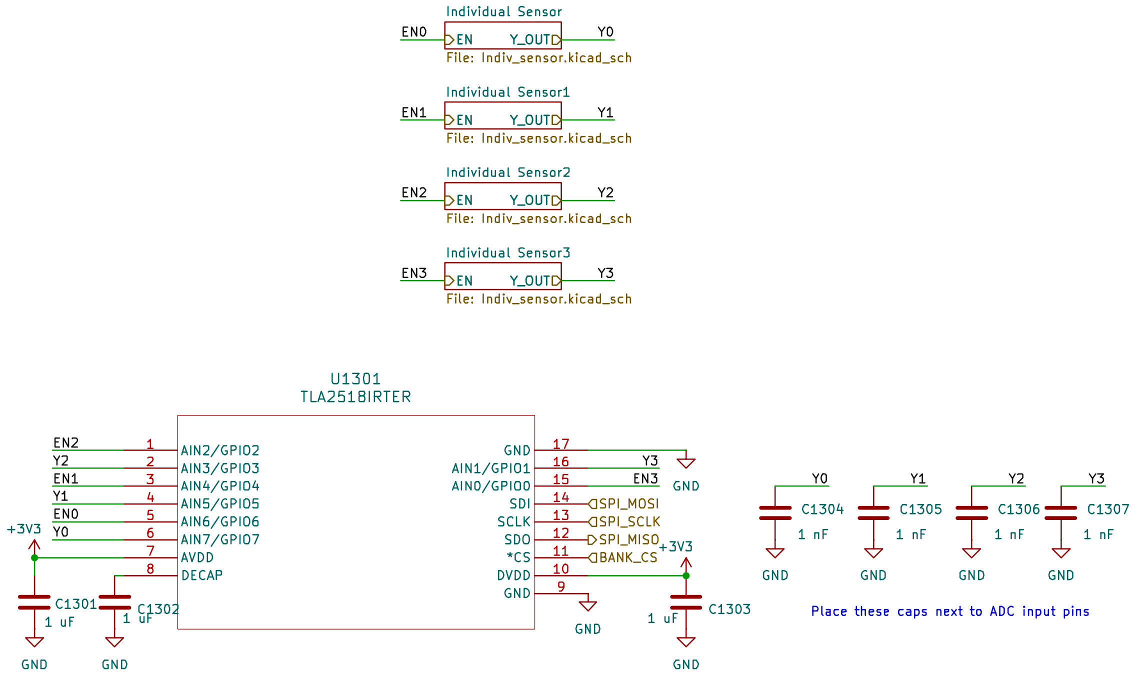
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 12/42



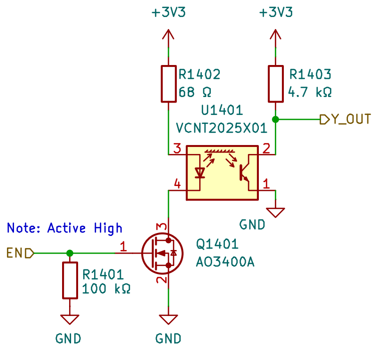
Sheet: /Sensor Array/4_Sensor_bank2/
File: 4_Sensor_Bank.kicad_sch

Title: 4 to 1 Channel Sensor Array

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 13/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

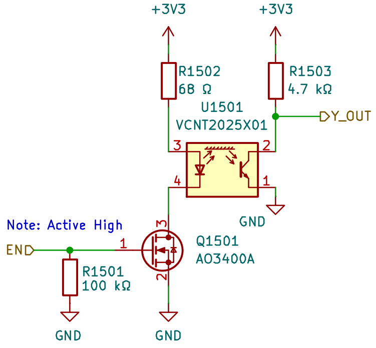
Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 14/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

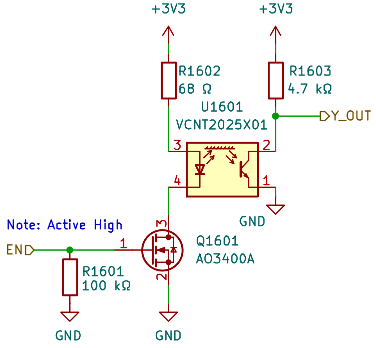
Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02		**Rev: C**
KiCad E.D.A. 9.0.7			Id: 15/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

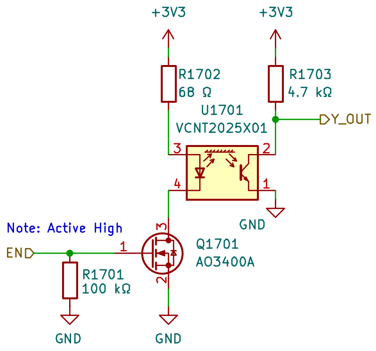
Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 16/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



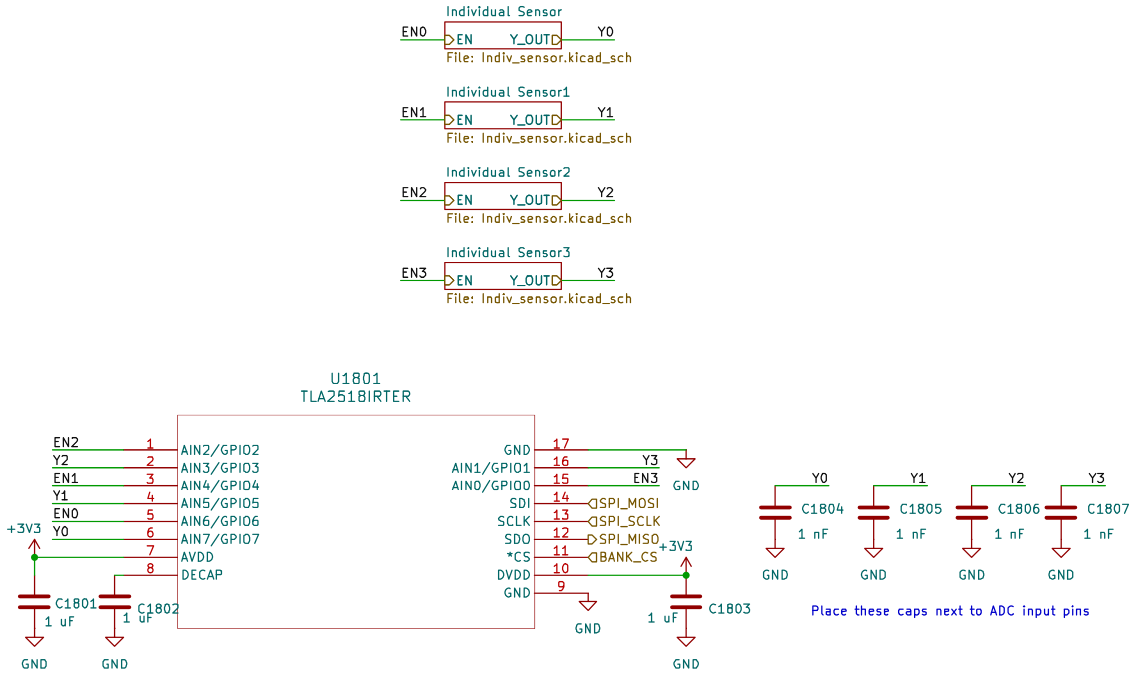
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 17/42



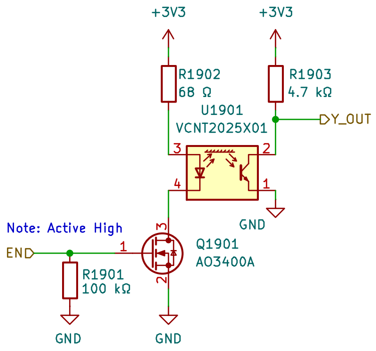
Sheet: /Sensor Array/4_Sensor_bank3/
File: 4_Sensor_Bank.kicad_sch

Title: 4 to 1 Channel Sensor Array

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 18/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

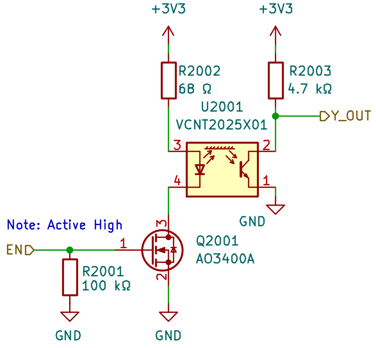
Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 19/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

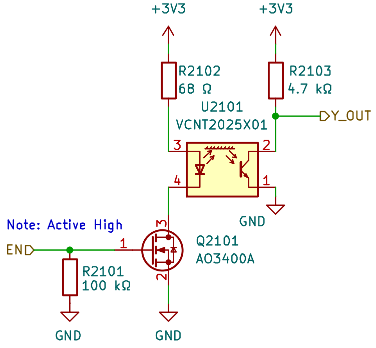
Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 20/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

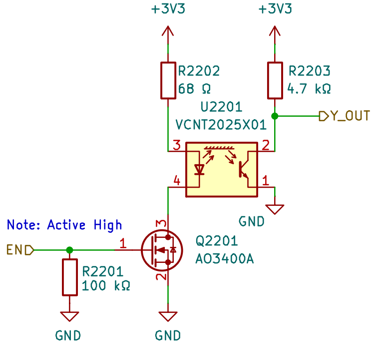
This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor2/	
File: Indiv_sensor.kicad_sch	
Title:	
Size: A4 — Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7	Id: 21/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation

Note: Active High



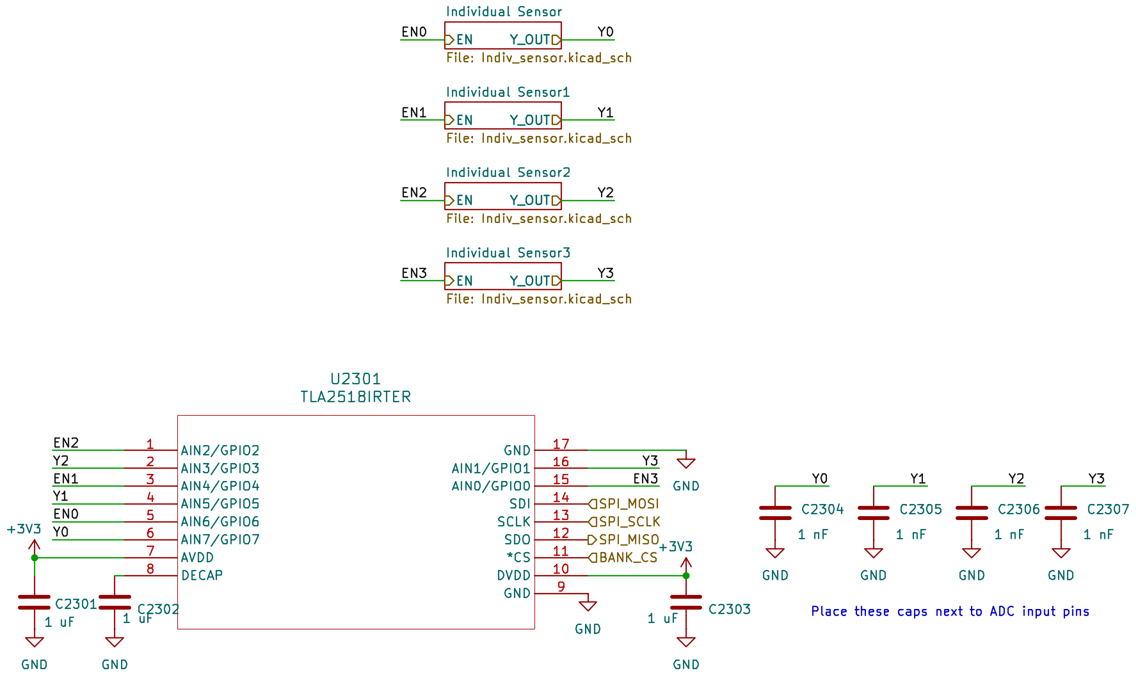
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 22/42



Sheet: /Sensor Array/4_Sensor_bank4/
File: 4_Sensor_Bank.kicad_sch

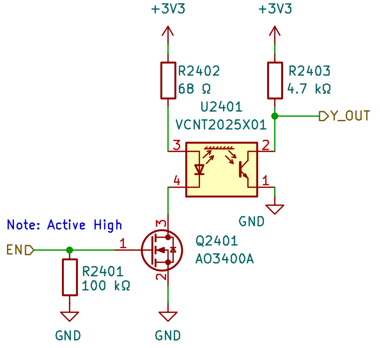
Title: 4 to 1 Channel Sensor Array

Size: A4	Date: 2026-01-02		Rev: C
KiCad E.D.A. 9.0.7			Id: 23/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation

Note: Active High



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

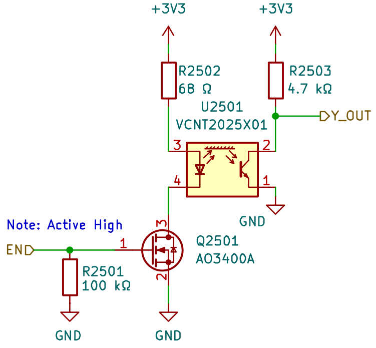
Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 24/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

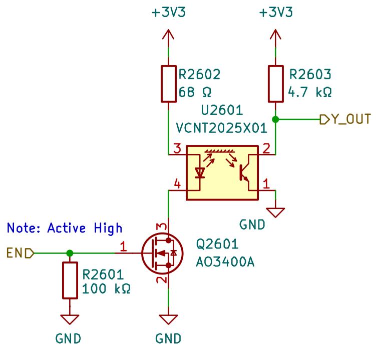
Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 25/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

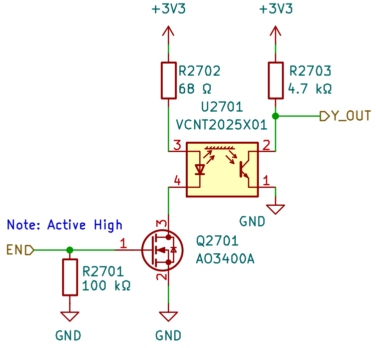
Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 26/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

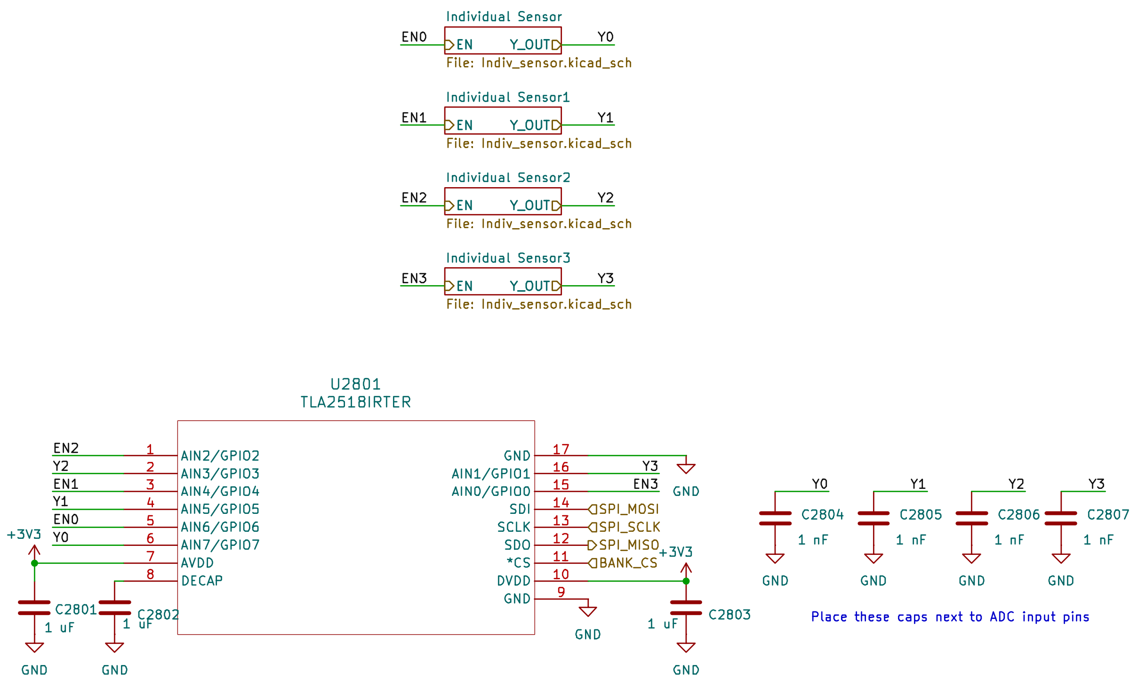
This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4 | Date: 2026-01-02 | **Rev: C**

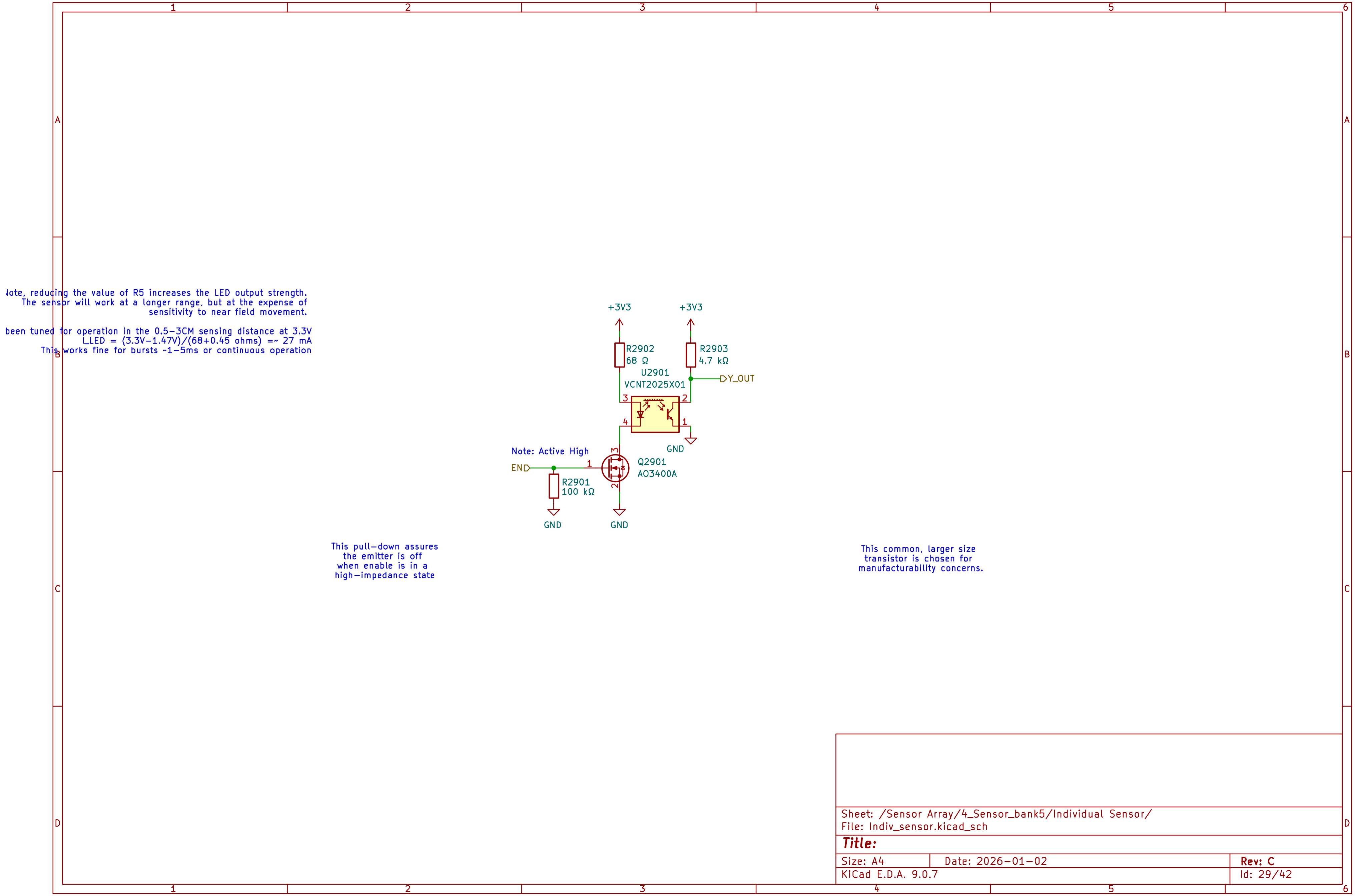
KiCad E.D.A. 9.0.7 | Id: 27/42



Sheet: /Sensor Array/4_Sensor_bank5/
File: 4_Sensor_Bank.kicad_sch

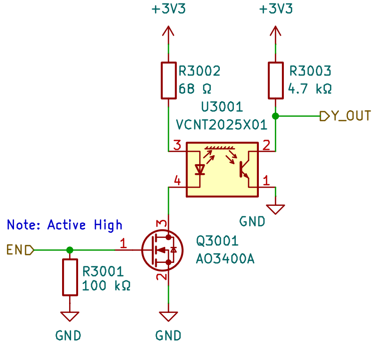
Title: 4 to 1 Channel Sensor Array

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 28/42


ote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.
been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation
+3V3
+3V3
R2902
68 Ω
R2903
4.7 kΩ
U2901
VCNT2025X01
Y_OUT
GND
Note: Active High
EN
Q2901
AO3400A
R2901
100 kΩ
GND
GND
This pull-down assures
the emitter is off
when enable is in a
high-impedance state
This common, larger size
transistor is chosen for
manufacturability concerns.
Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor/
File: Indiv_sensor.kicad_sch
Title:
Size: A4
Date: 2026-01-02
Rev: C
KiCad E.D.A. 9.0.7
Id: 29/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor1/
File: Indiv_sensor.kicad_sch

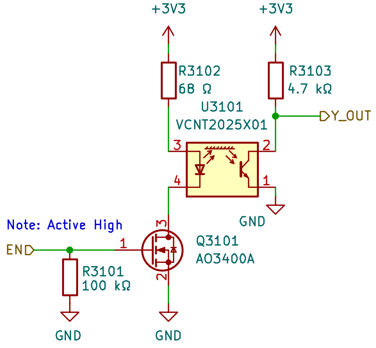
Title:

Size: A4 | Date: 2026-01-02 | **Rev: C**

KiCad E.D.A. 9.0.7 | Id: 30/42

ote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

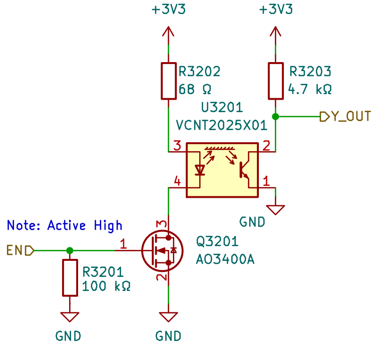
Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 31/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



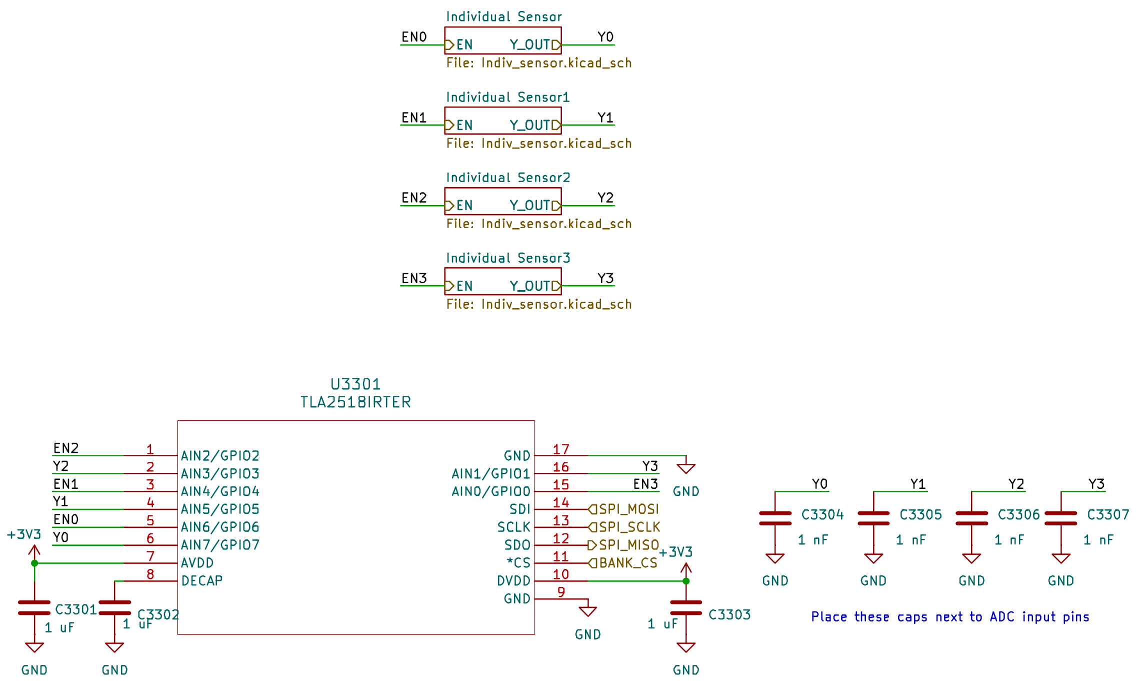
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

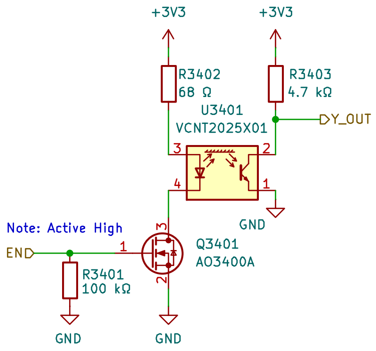
Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 32/42



Sheet: /Sensor Array/4_Sensor_bank6/	
File: 4_Sensor_Bank.kicad_sch	
Title: 4 to 1 Channel Sensor Array	
Size: A4 Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7	Id: 33/42

ote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

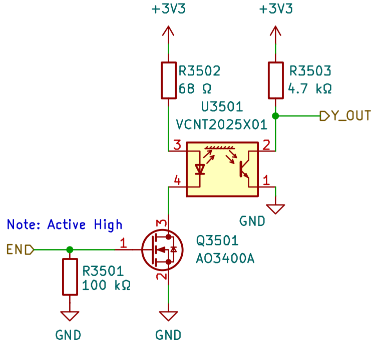
Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 34/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

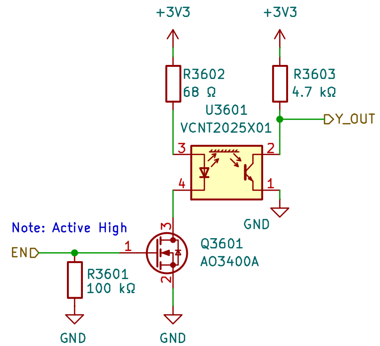
Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 35/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

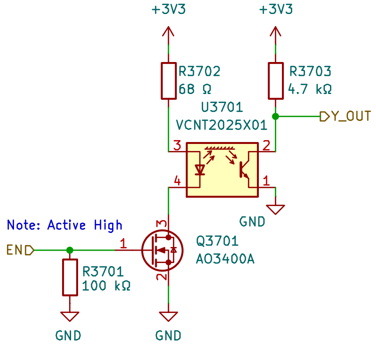
Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 36/42

lote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



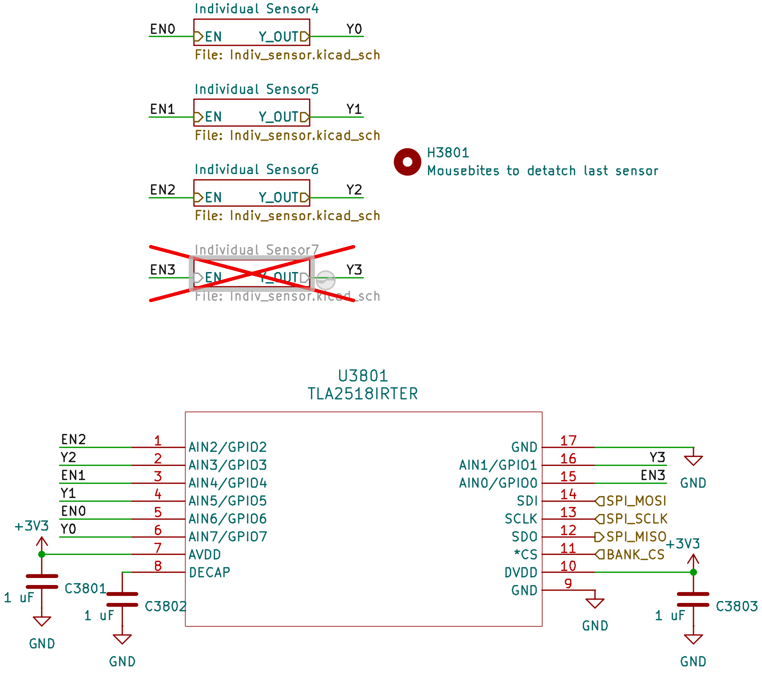
This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 37/42



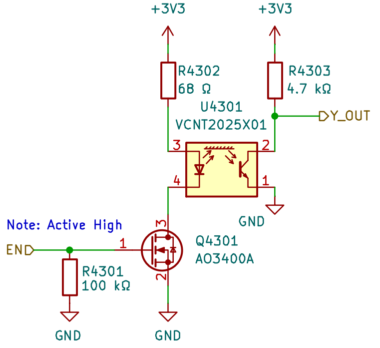
Sheet: /Sensor Array/Last_sensor_bank/
File: last_sensor_bank.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 38/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

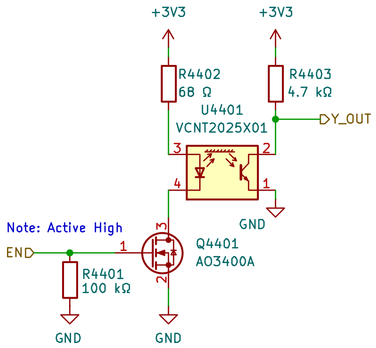
Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor4/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 43/42

ote, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

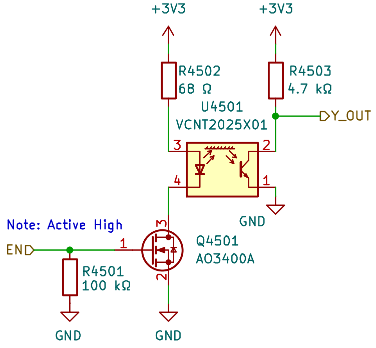
Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor5/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 44/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

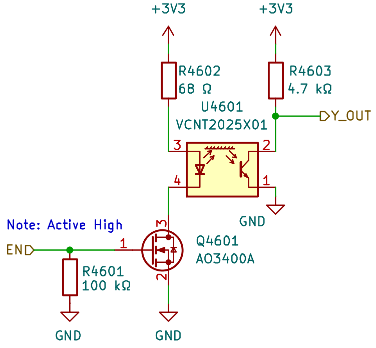
Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor6/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 45/42

Note, reducing the value of R5 increases the LED output strength.
The sensor will work at a longer range, but at the expense of
sensitivity to near field movement.

been tuned for operation in the 0.5-3CM sensing distance at 3.3V
I_LED = (3.3V-1.47V)/(68+0.45 ohms) =~ 27 mA
This works fine for bursts ~1-5ms or continuous operation



This pull-down assures
the emitter is off
when enable is in a
high-impedance state

This common, larger size
transistor is chosen for
manufacturability concerns.

Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor7/
File: Indiv_sensor.kicad_sch

Title:

Size: A4	Date: 2026-01-02	Rev: C
KiCad E.D.A. 9.0.7		Id: 46/42